

Quiz 3: Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. You may use the limit laws and examples covered in Lectures 3 and 4. When a proof directly from the definition is requested, use the epsilon–delta definition. All functions are real-valued. Unless otherwise specified, a function whose limit at a is considered is defined on an open interval containing a , except possibly at a itself. Conditions on x refer to points in the relevant domain, or common domain when functions are combined.

1. Find

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}.$$

Explain why the quotient law cannot be applied directly to the original numerator and denominator, and justify your calculation.

Solution.

The numerator and denominator both tend to zero. The quotient law requires a nonzero limit for the denominator, so it cannot be applied directly to the original expression. For $x \neq 1$, factor and cancel to obtain

$$\frac{x^3 - 1}{x - 1} = \frac{(x - 1)(x^2 + x + 1)}{x - 1} = x^2 + x + 1.$$

The two expressions agree for every $x \neq 1$, so they have the same limit at 1. The polynomial limit laws give

$$\boxed{\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = 1 + 1 + 1 = 3.}$$

2. Define

$$f(x) = \begin{cases} x + 1, & x < 0, \\ 0, & x = 0, \\ 3 - x, & x > 0. \end{cases}$$

Prove that f has no finite limit at zero. For each candidate L , use the negated definition to give a fixed $\varepsilon_0 > 0$ and a suitable x for every $\delta > 0$.

Solution.

Fix any candidate $L \in \mathbb{R}$. Take $\varepsilon_0 = 1/2$. For an arbitrary $\delta > 0$, let

$$h = \min\{\delta/2, 1/4\}.$$

Then $0 < h < \delta$ and $h \leq 1/4$.

If $L \leq 2$, choose $x = h > 0$. Since $f(h) = 3 - h$,

$$f(h) - L = 3 - h - L \geq 1 - h \geq \frac{3}{4} > \varepsilon_0.$$

Thus $|f(h) - L| \geq \varepsilon_0$.

If $L > 2$, choose $x = -h < 0$. Since $f(-h) = 1 - h$,

$$|f(-h) - L| = L - 1 + h > 1 > \varepsilon_0.$$

In either case, there is an x with

$$0 < |x| < \delta \quad \text{and} \quad |f(x) - L| \geq \varepsilon_0.$$

This holds for every $\delta > 0$, with ε_0 fixed in advance. The negated definition rules out this L . Since L was arbitrary, f has no finite limit at zero.